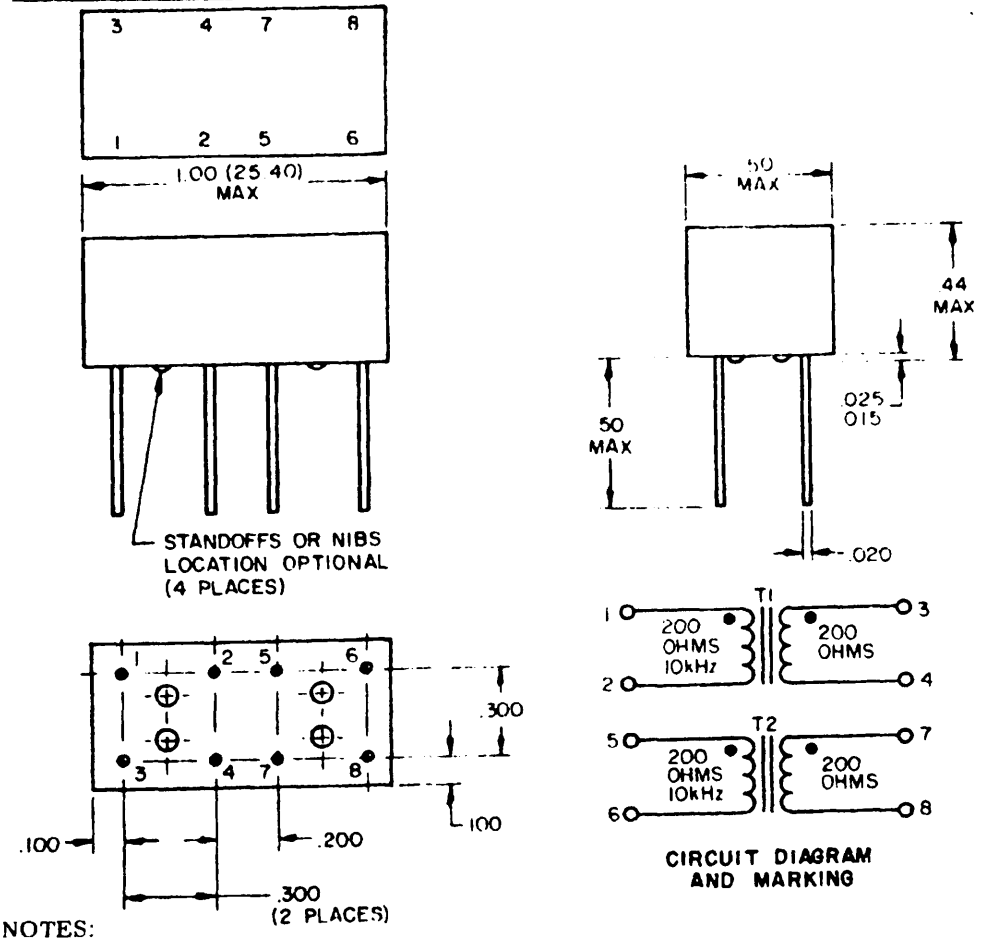
MILITARY SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY, TF5V21ZZ318

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Marking and terminal location shall be on top of the case.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical rating:

Source impedance (1-2): 200 ohms, 10 kHz.

Load impedance (3-4): 200 ohms.

Source impedance (5-6): 200 ohms, 10 kHz.

Load impedance (7-8): 200 ohms.

Primary power level - 3.6 milliwatts.

Working voltage - 55 volts, 10 kHz.

Design, configuration, and physical dimensions: See figure.

Material: Epoxy.

Weight: 8 grams.

Duty cycle: Continuous.

Terminal: Pin, N-2 type, in accordance with MIL-STD-1276.

Temperature range: -55°C to 155°C.

Terminal strength: Method 211, MIL-STD-202, test condition A, 2 pounds.

Dielectric withstanding voltage: Method 301 of MIL-STD-202; test voltage, 300 volts, rms.

Electrical characteristics:

Primary impedance - (1-2), (5-6): 200 ohms, 10 kHz.

Secondary load impedance - (3-4), (7-8): 200 ohms.

No load voltage ratio (1-2): Apply 0.85 volts rms, 10 kHz.
(3-4): Measure 0.85 volts, rms, ±1 percent.
(5-6): Apply 0.85 volts, rms, 10 kHz.
(7-8): Measure 0.85 volts rms, ±1 percent.

Excitation current (1-2): Apply 0.85 volts rms, 10 kHz. Measure 2.0 milliamperes rms maximum.
(5-6): Apply 0.85 volts rms, 10 kHz. Measure 2.0 milliamperes rms maximum.

DC resistance at 20°C: Primary (1-2) or (5-6) - 6.2 ohms maximum.
Secondary (3-4) or (7-8) - 5.9 ohms maximum.

Leakage inductance: Apply 0.05 volts rms, 10 kHz across (3-4) or (7-8).
Measure 3.0 μHy maximum across (1-2) or (5-6).

Polarity - Additive, with terminals (2-3) and (6-7) connected.

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Temperature rise: 10°C.

Vibration, high frequency: Method 204, MIL-STD-202, test condition A.

Shock (specified pulse): Method 213, MIL-STD-202, test condition I.

Salt spray: Test in accordance with Method 101 of MIL-STD-202.

Part number: M27/32-01.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MU, WC, SG, MF
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0500-2)

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To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-27/32 TRANSFORMER, AUDIO FREQUENCY, TF5V21ZZ318

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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